

I build systems and compiler components in C++23/C17/Rust: LLVM passes, memory-dependence analysis, CFG/SSA, graph algorithms, x86-64 code generation, and reproducible test oracles.

LLVM / C++23

Optimization pass, graph algorithms, CMake, clang-tidy, ASan/UBSan

1st of 49 · 104/104

Conservative PS-form memory-dependence analysis with an exact oracle

x86-64 pipeline

SSA-like IR, liveness, linear scan, iced-x86, and differential validation

EXPERIENCE		SKILLS
July 1 — August 31, 2026	MCST — compiler engineering intern Developing an LLVM optimization pass, implementing tested graph algorithms, and working with CFG transformations.	C++ / toolchain C++23, C17, CMake, Linux, clang-tidy, ASan/UBSan, GitHub Actions
2026	PS-form Memory Dependence Analyzer - Implemented conservative analysis of parametric memory accesses; ranked 1st of 49 with 104/104 tests. - Built an exact oracle for small domains, metamorphic tests, and reproducible WA/TL counterexamples.	Compiler analysis CFG, dominance, SSA, SCC, dataflow, memory dependence, optimization passes
2026	x86-64 Codegen & Register Allocation Lab Designed SSA-like IR, liveness, linear scan, and parity checks between an IR interpreter and generated machine code.	Code generation x86-64, iced-x86, liveness, linear scan, IR interpretation, isolated execution Verification exact oracles, differential/metamorphic tests, stress testing, counterexample reduction
PROJECTS		EDUCATION
PS-form Memory Dependence Analyzer Ranked 1st of 49 with 104/104 tests, with reproducible wrong-answer and time-limit counterexamples.		Software Engineering student at HSE University
x86-64 Codegen & Register Allocation Lab A measurable pipeline with spill/reload metrics, code size, and reproducible result comparison.		RECOGNITION
UniversalToolchain / Wist2 1,465/1,465 tests with zero failures across nine packages and clean-consumer projects, plus interpreter/CIL parity and reproducible configuration checks.		The LangDev 2026 Program Committee accepted a talk on the architecture of UniversalToolchain/Wist2 for presentation in Torremolinos, Spain. Moscow / remote